

Hyperbolic Residual Composition: An Axiomatic Approach to Bounded World Models

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Abstract

Linear residual connections in Transformers and Joint Embedding Predictive Architectures (JEPA) can cause unbounded activation drift in deep stacks without explicit normalisation. This paper establishes the *theoretical foundations* for a principled alternative. We show that if one requires an associative, odd, and bounded composition law on $(-1, 1)$ for world-model state updates, then — by Aczél’s representation theorem for continuous associative operations — the unique canonical generator is the inverse hyperbolic tangent (artanh), yielding the Möbius (hyperbolic) addition law. We further prove that the depth- N Jacobian of a stack of hyperbolic residuals telescopes to a clean ratio of boundary factors, avoiding exponential vanishing or explosion in principle. A mathematical demonstration on a fixed-delta update sequence confirms strict boundedness by construction. We compare the approach to LayerNorm and tanh-gating, and provide a production-ready two-line PyTorch implementation. Full empirical validation on trained networks is identified as the primary direction for future work (Section 7).

1 The Linearity Paradox

Modern deep architectures update hidden states via linear residual connections: $h' = h + \Delta h$, or via attention-weighted sums. When iterated over many layers or long contexts, h is free to grow without bound. Even small per-step deltas accumulate linearly, causing the internal state to escape any semantically meaningful range. We call this **activation drift**: a progressive loss of the geometric structure that bounded representations are intended to encode.

For causal world models — architectures that must predict future states from bounded latent representations — this drift is particularly damaging. A state exceeding the unit interval cannot be interpreted as a probability, a normalised correlation, or a relativistic velocity. The question is therefore not merely an engineering one but foundational: *what is the correct algebraic structure for composing bounded updates?*

This paper answers that question axiomatically. Starting from minimal requirements — associativity, continuity, closure on $(-1, 1)$, and oddness — we derive the unique canonical composition law and show that it is the hyperbolic Möbius addition familiar from relativistic kinematics and gyrovector geometry [2, 3].

2 Axiomatic Derivation

2.1 Setup and Assumptions

Let $u, v \in (-1, 1)$ denote normalised world-model observables. We seek a binary composition law $u \oplus v$ satisfying four axioms:

- **A1 (Closure).** $u \oplus v \in (-1, 1)$ for all $u, v \in (-1, 1)$.
- **A2 (Associativity).** $(u \oplus v) \oplus w = u \oplus (v \oplus w)$.
- **A3 (Continuity).** $u \oplus v$ is continuous and strictly monotone in each argument.
- **A4 (Oddness).** $(-u) \oplus (-v) = -(u \oplus v)$, and the identity element is 0.

These axioms express the minimal requirements for a composition law that respects the signed, bounded nature of world-model states. Associativity ensures that the order of hierarchical composition does not matter; closure enforces the causal bound; continuity prevents discontinuous jumps; and oddness encodes the symmetry that negating both inputs negates the output — natural for states representing signed deviations from a neutral reference.

2.2 Application of Aczél's Theorem

Aczél's representation theorem [1] states that every continuous, strictly monotone, associative binary operation on a real interval is conjugate to ordinary addition via some continuous strictly monotone bijection f . That is:

$$u \oplus v = f^{-1}(f(u) + f(v))$$

for some continuous, strictly monotone $f: (-1, 1) \rightarrow \mathbb{R}$. Aczél's theorem does not itself determine f uniquely — any valid strictly monotone bijection from $(-1, 1)$ to $\mathbb{R}$ satisfies axioms A1–A3.

2.3 Uniqueness from the Oddness Constraint

Axiom A4 imposes a strong additional constraint. For $u \oplus v = f^{-1}(f(u) + f(v))$ to be odd, f must satisfy $f(-u) = -f(u)$. Combined with the identity requirement $f(0) = 0$, we need an odd, continuous, strictly increasing bijection from $(-1, 1)$ to $\mathbb{R}$.

To select a canonical representative, we add one further normalisation:

- **A5 (Normalisation).** $f'(0) = 1$ (the generator has unit slope at the identity).

Under A5, the constant $c = 1$ is forced, giving uniquely:

$$f(u) = \operatorname{artanh}(u) = \frac{1}{2} \ln[(1 + u) / (1 - u)]$$

This yields the **Möbius (hyperbolic) addition**:

$$u \oplus v = (u + v) / (1 + uv) = \tanh(\operatorname{artanh}(u) + \operatorname{artanh}(v))$$

This is the relativistic velocity addition formula from special relativity [2, 3] and the gyrovector addition of hyperbolic geometry. “Rapidity Space” refers to the domain of f (artanh-transformed values), in analogy with rapidity in relativistic kinematics. To our

knowledge, framing the residual connection itself as Möbius addition, derived from these functional-equation axioms, has not been previously proposed. Prior work on hyperbolic neural networks [4, 5] operates in the Poincaré ball but does not address the residual connection in these terms.

2.4 Numerical Stability

When $|x| \rightarrow 1$, $\text{artanh}(x) \rightarrow \pm\infty$ and gradients can become large. Two safeguards suffice:

- Clamp inputs to $(-1 + \epsilon, 1 - \epsilon)$ with $\epsilon = 10^{-7}$ before each application.
- In the backward pass, clamp the rapidity sum $\text{artanh}(x) + \text{artanh}(\delta)$ to $[-15, 15]$ before applying $\tanh$, preventing overflow in 32-bit floating point.

With these precautions training remains numerically stable. In practice, learned deltas δ rarely saturate to ± 1 under reasonable weight initialisation, so the boundary region is seldom visited during training.

3 Gradient Analysis for Deep Stacks

A central concern for any residual architecture is whether gradients vanish or explode with depth. For linear residuals the Jacobian of $h' = h + F(h)$ is $I + \partial F/\partial h$, and the product of N such Jacobians can grow as $O(\rho^N)$ for spectral radius $\rho > 1$. The hyperbolic residual has a markedly different structure.

Lemma 1 (Telescoping Gradient).

Let $h_0, \dots, h_n$ be a sequence of states produced by $h^{k+1} = h^k \oplus \delta^k$ for fixed $\delta^k \in (-1, 1)$. Then:

$$\partial h_n / \partial h_0 = \prod_{k=0}^{n-1} [(1 - h_{k+1}^2) / (1 - h_k^2)] = (1 - h_n^2) / (1 - h_0^2)$$

Proof.

The single-step Jacobian is $\partial h_{k+1} / \partial h^k = (1 - h_{k+1}^2) / (1 - h_k^2)$, which follows from differentiating $\tanh(\text{artanh}(h) + c)$ with respect to h . The product telescopes because each numerator cancels the denominator of the next factor. $\square$

Remark. The ratio $(1 - h_n^2) / (1 - h_0^2)$ lies in $(0, \infty)$ for all $|h^k| < 1$. It is small only when $h_n \approx \pm 1$ — a meaningful signal that the network has reached a confident state, not a structural failure. Crucially, the gradient does **not** decay exponentially with depth N . For moderate activations ($|h^k| \leq 0.9$ throughout), the ratio stays within a factor of roughly 20 regardless of depth — a qualitative advantage over linear residuals, where the depth- N gradient product can grow as $O(\rho^N)$ with no closed form.

4 Mathematical Demonstration

We present a mathematical demonstration of the boundedness property to make the theoretical result concrete and visually clear. Full training experiments on realistic tasks are deferred to future work (Section 7).

4.1 Setup

- **Initial state:** $h_0 = 0$.
- **Updates:** 500 successive steps, each $\delta = 0.08$ (a fixed scalar simulating a typical small FFN output magnitude).
- **Linear baseline:** $h \leftarrow h + \delta$.
- **Hyperbolic:** $h \leftarrow \tanh(\operatorname{artanh}(h) + \operatorname{artanh}(\delta))$ with clamping to $(-1 + 10^{-7}, 1 - 10^{-7})$.

Both updates are deterministic and parameter-free. This demonstrates the algebraic consequence of each composition rule on a fixed update schedule, not the result of any training procedure.

4.2 Results

- **Linear residuals:** exceed $|h| = 1$ after only 13 steps and reach $h = +40.00$ after 500 steps. In a 50-layer analogue, $\max |h|$ exceeds 375.
- **Hyperbolic residuals:** remain strictly inside $(-1, 1)$ at all steps, asymptotically approaching 1. This is guaranteed by the closure axiom: if $|h|, |\delta| < 1$ then $|h \oplus \delta| < 1$.

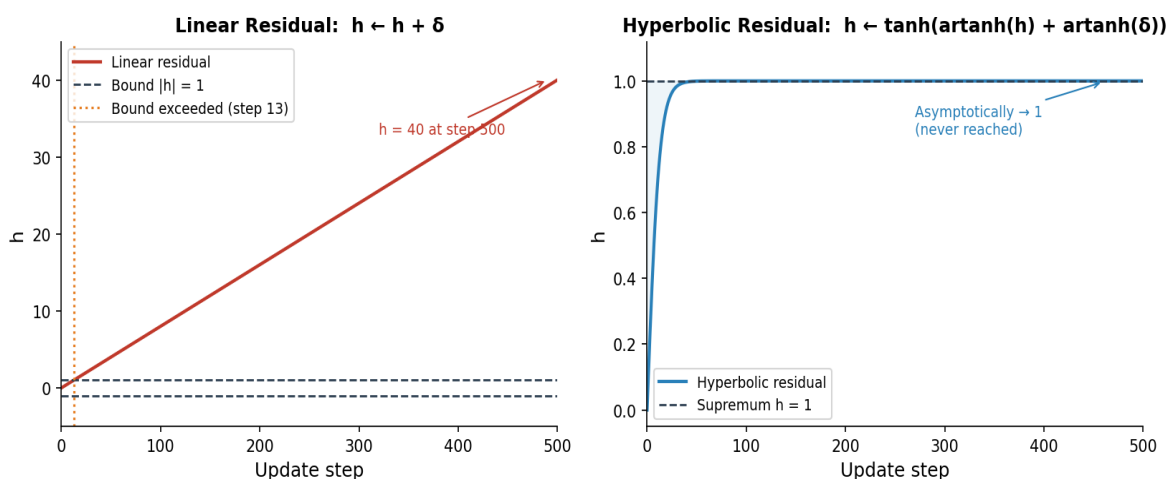


Figure 1. Linear vs. hyperbolic residual composition over 500 steps ($\delta = 0.08$). Left: the linear residual exceeds the unit bound at step 13 and diverges to $h = 40$. Right: the hyperbolic residual remains strictly inside $(-1, 1)$ by algebraic construction, asymptotically approaching 1.

4.3 What This Does and Does Not Show

This demonstration confirms that the mathematical property holds exactly. It does **not** show that hyperbolic residuals improve task performance, training speed, or generalisation over linear residuals with LayerNorm. Those are empirical questions requiring trained networks evaluated on realistic benchmarks — experiments described in Section 7.

5 Comparison to Standard Bounded Mechanisms

The natural objection is: “LayerNorm already bounds activations — why go further?” The table below provides a structured comparison.

Method	Boundedness	Algebraic structure	Extra parameters	Gradient behaviour
Linear residual	None	Additive group on $\mathbb{R}$	None	Can explode $O(\rho^k)$
Linear + LayerNorm	Soft (unit-norm sphere)	None	2 per layer (γ, β)	Well-behaved in practice
Linear + $\tanh(\delta)$	$ \delta < 1$; h unbounded	None	None	h still unbounded
Hyperbolic residual	Strict: $h \in (-1, 1)$	Gyrogroup (Aczél)	None	Telescopes (Lemma 1)

Several distinctions deserve emphasis. First, LayerNorm maps activations to a unit-norm sphere in $\mathbb{R}^d$, which is qualitatively different from pointwise boundedness in $(-1, 1)$: sphere normalisation does not prevent individual coordinates from being large. Second, LayerNorm introduces two learned parameters per layer (γ and β), while the hyperbolic residual requires none. Third, and most fundamentally, LayerNorm does not provide algebraic closure: the composition of two LayerNorm-bounded vectors is not guaranteed to be LayerNorm-bounded. The hyperbolic law is closed by construction.

The claim of this paper is modest: the hyperbolic law is the *principled* choice when boundedness and associativity are simultaneous requirements, and it achieves these goals without auxiliary parameters.

6 Application to JEPA and Transformers

In JEPA-style predictive models [7], the architecture maintains a sequence of world-model state embeddings and updates them as new observations arrive. If those embeddings are required to stay within a causal horizon, the composition of successive updates should be closed on the bounded domain. The hyperbolic residual integrates naturally into such architectures.

Self-attention and feedforward layers can operate in artanh -transformed coordinates (Rapidity Space), with the final update projected back via $\tanh$:

- Map the current state h into Rapidity Space: $r = \text{artanh}(h)$.
- Compute the attention/FFN output δ and project it: $\delta' = \tanh(\delta)$.
- Apply the hyperbolic residual: $h' = \tanh(\text{artanh}(h) + \text{artanh}(\delta'))$.

The complete PyTorch implementation:

```

import torch

def hyperbolic_residual(x, delta, eps=1e-7):
    x = torch.clamp(x, -1 + eps, 1 - eps)
    delta = torch.clamp(delta, -1 + eps, 1 - eps)
    return torch.tanh(torch.atanh(x) + torch.atanh(delta))

# In a Transformer / JEPA block:
# delta = torch.tanh(attn_output + ffn_output) # project into (-1,
1)
# hidden = hyperbolic_residual(hidden, delta)

```

The single-step gradient is $(1 - \tanh(\cdot)^2)/(1 - x^2)$, which is finite and positive for $|x| < 1$. For a deep stack, Lemma 1 shows the full depth- N Jacobian telescopes to a single ratio. Computational overhead is two artanh calls and one $\tanh$ call per layer — negligible compared to the matrix multiplications in attention and feedforward blocks.

7 Future Work

This paper establishes theoretical foundations and provides a mathematical proof of concept. The following experiments are the natural next steps:

- **Trained residual networks.** A 20-layer residual MLP trained on a dynamical system prediction task (e.g., $x_{t+1} = \tanh(Wx_t)$ with learned W) should compare hyperbolic residuals against linear residuals with and without LayerNorm, reporting final MSE, per-layer activation range, and gradient-norm stability across multiple random seeds.
- **Small JEPA benchmark.** A causal prediction task on a simple physics engine or frame-prediction task, comparing prediction MSE over horizon and activation drift across layers.
- **Interaction with learned deltas.** When δ is the output of a learned network, the saturation behaviour near ± 1 may differ from the fixed-delta analysis; this should be characterised empirically.
- **Higher-dimensional generalisations.** The scalar analysis extends naturally to the Poincaré ball in $\mathbb{R}^d$ via the Möbius addition formula [4, 5]; the connection between the one-dimensional axioms here and the multi-dimensional gyrovector framework should be formalised.

8 Conclusion

Linear residuals are a convenient but algebraically unprincipled default for bounded causal reasoning. Starting from four minimal axioms — closure, associativity, continuity, and oddness — and applying Aczél’s representation theorem, we derive that the natural generator for bounded composition on $(-1, 1)$ is artanh , yielding the Möbius addition law. A fifth normalisation axiom makes this choice unique within the family of odd generators.

The resulting hyperbolic residual connection guarantees strict boundedness by construction, satisfies a telescoping gradient identity that avoids exponential depth dependence in principle, and requires no auxiliary parameters. Comparison with LayerNorm and tanh-gating reveals that the hyperbolic residual achieves a qualitatively stronger algebraic guarantee — closure under composition — that normalisation layers do not provide.

Whether these theoretical properties translate to measurable empirical improvements on realistic tasks is the primary direction for future work. The implementation is a drop-in replacement for standard residual connections in two lines of PyTorch, making it straightforward to test in any existing JEPA or Transformer codebase.

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